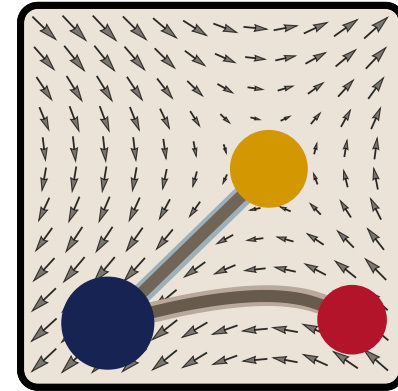




Mathematical modelling in ecology and evolution (EEB314/1430)

See <https://osmond-lab.github.io/eeb314> for more info

Mathematics is central to science because it provides a rigorous way to go from a set of assumptions to their logical consequences. In ecology & evolution this might be how we think a virus will spread and evolve, how climate change will impact a threatened population, or how much genetic diversity we expect to see in a randomly mating population. In this course you'll learn how to build, analyze, and interpret mathematical models of increasing complexity through readings, lectures, labs, and a final project. Our focus is on deterministic dynamical models (recursions and differential equations), which requires us to use and learn some calculus and linear algebra.

Please see the University of Toronto Academic Calendar for more details on the course prerequisites and additional information on the distribution/breadth requirements this course satisfies.

Next taught: Fall 2026

Previously taught: Fall 2025, Fall 2024, Fall 2022 (EEB430), Fall 2021 (EEB430)

Instructors

Professor

Matthew Osmond (he/him)

- email: mm.osmond@utoronto.ca
- office: Earth Sciences (ES) room 3041
- office hour: Wednesday 2pm
- website: osmond-lab.github.io

Teaching assistant

Mete Yukel (they/he)

- email: mete.yukel@mail.utoronto.ca
- office: Earth Sciences (ES) room 3047
- office hour: Thursday 1pm

When and where

Lectures

- Monday 10:10 - 11:00 AM, Earth Sciences (ES) room 3087
- Wednesday 10:10 - 11:00 AM, Earth Sciences (ES) room 3087

Labs

- Wednesday, 3:10 - 5:00 PM, Earth Sciences (ES) room 3087

Course structure

Learning objectives

- ☒ build a model: go from a verbal description of a biological system to a set of equations
- ☒ analyze a model: manipulate a set of equations into a mathematical expression of interest
- ☒ interpret a model: translate mathematical expressions back into biological meaning

Weekly tasks

- ☒ attend two lectures (2h)
- ☒ participate in one lab (2h)
- ☒ do practice problems (listed at end of each lecture)
- ☒ read assigned portions of the text (when more info needed)

Grading scheme

- in-class tests: 4 x 20% (4 x 15% for grad students)
- final project: 20% (40% for grad students)

Textbook

Otto & Day 2007. A biologist's guide to mathematical modeling in ecology and evolution.

Final project

Construct your own model

In this project you will use the tools you've learned in class and apply them to a model that you develop. The model can be about any phenomenon in ecology and evolution, as long as you make up the model. I suggest you stick with dynamical models and analyses learned in class.

You'll do the final project in two parts.

- ☒ Part 1 (25%)
 - State your biological question
 - Describe your model in words (ie, the main assumptions) and, optionally, with a diagram (eg, flow diagram or life cycle)
 - Write down the equations that you plan to analyze
 - Describe what analysis you plan to do and how this will answer your question
 - Max 500 words
- ☒ Part 2 (75%)
 - Motivate and state your biological question
 - Describe your model assumptions in detail, defining all parameters and variables
 - Write down the equations for your model
 - Analyze your model
 - Explain how the results address your original question
 - Suggest how the model could be improved or extended
 - Plots and supplementary Jupyter notebooks welcome but not required
 - Max 2000 words (4000 for grad students)

Class presentations

Grad students will also present their final project in class. Everyone may be asked to give a 2m lightning presentation on part 1.

Tip

If you are having trouble coming up with a new model, take one of the models that we've analysed in the course and change one or more of its underlying assumptions to get a new set of equations. Then analyse these equations. Discuss the differences between the assumptions used and also between the results obtained.

AI policy

Your final project must be conducted and submitted individually. That said, you are encouraged to discuss your ideas and analyses with others as a way to shape and improve them. Your submitted work must, however, reflect your own understanding and conclusions. Similarly, use of artificial intelligence (AI) tools, such as ChatGPT, Claude, Copilot, DeepL, Gemini, Google Translate, Grammarly, Llama, etc., is allowed only to the extent that it aids your learning and understanding. Any use of AI must cite the tool using APA guidelines and specify exactly how AI was used, including the search term, the chat history link (if available), and how the results were used. In cases where a chat history is not available, please save a copy of the chat to your computer in case we ask for more details. Again, your submitted work must reflect your own understanding and conclusions. Getting AI help to learn how to improve your grammar or to debug code or to check an answer is appropriate in this course (if cited); having AI solve problems for you or suggest ideas for a course project would not be acceptable as then your submitted work cannot be your own. If in doubt, please ask!